Google Cloud H2S

AI Builder Cup

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Team Details

a. Team name: **AquaSense AI**

b. Team leader name: **Kartik Batta**

c. Problem Statement: Water allocation under drought — deciding where to measure and how to split a fixed water budget across farm zones to minimise crop stress

Brief about the idea

AquaSense is a decision-support engine for irrigation authorities who must ration water during a drought. Given a hard water budget (kL), a small field-sensor quota (K measurements) and uncertain water-need estimates for each farm zone, it decides which zones are worth measuring and how to split the water so that weighted crop stress is as low as possible — without ever exceeding the supply cap.

1. Model uncertainty

Each zone carries a prior need μ ± σ, a crop risk weight w (1.0–4.5), soil type, NDVI and telemetry status.

2. Decide where to sense

A Value-of-Information engine Monte-Carlo-simulates (25 draws per zone) how measuring each zone would change the optimal allocation.

3. Allocate under a hard cap

A convex water-filling optimizer minimises Σ w·(need − allocation)² subject to Σ allocation ≤ budget.

4. Explain it

A Gemini copilot explains the decision in 4 languages, quoting only numbers produced by the simulator.

Prototype ships with three Indian river-basin scenarios — Indus-Ganges heatwave (35% supply cut), Mahanadi drought (50% reservoir drawdown), Deccan monsoon delay — plus a randomized scenario generator.

Opportunities

How is it different from existing ideas?

Typical tools either split water in proportion to estimated demand, or send sensors wherever uncertainty is highest. AquaSense treats sensing as part of the decision: a measurement is scored by how much it is expected to shift the constrained allocation and cut weighted crop-stress loss. Every run is benchmarked live against three baselines on the same scenario and hidden ground truth.

How will it solve the problem?

During a shortfall, authorities otherwise split a fixed volume blind. Spending a small sensor quota and then optimizing the split cut the simulated crop-stress loss proxy by 25–38% vs. demand-proportional allocation across the three basin scenarios, with 0 kL budget overdraft in every run — and turns the result into canal-gate schedules a farmer cooperative can follow.

USP of the proposed solution

Sensing + hard-constraint allocation + accountability: a Gemini copilot guardrailed to quote only simulator numbers (with a deterministic offline fallback), an exportable JSON audit log, and three persona views — director, agronomist, farmer co-op — in English, Hindi, Spanish and Swahili with voice.

List of features offered by the solution

Four-policy comparison matrix
Demand-proportional, uncertainty-weighted, max-uncertainty sensing and AquaSense VOI, run on identical constraints.

VOI sensing engine
Ranks zones by expected loss reduction; flags tipping points via allocation elasticity.

Constrained allocation optimizer
Water-filling by bisection on the Lagrange multiplier; exact budget preservation.

KPI scorecard
Crop-stress loss proxy, Jain's fairness index, budget compliance, VOI advantage.

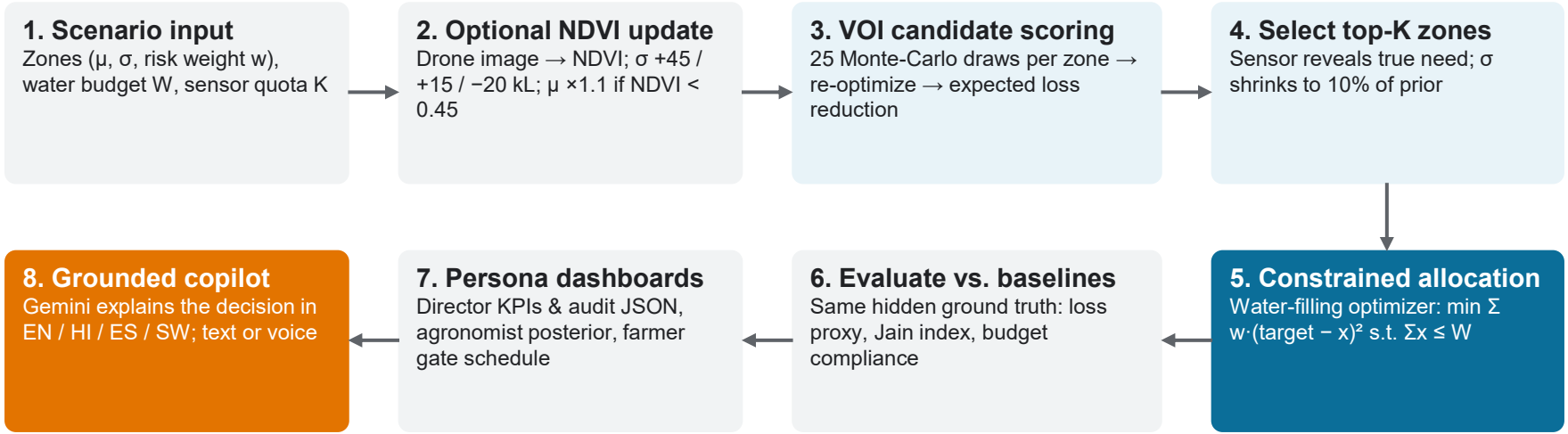
GIS basin grid
Per-zone allocation vs. prior, satisfaction, ±σ halo, telemetry & battery status.

Drone / NDVI canopy scanner
Preset or uploaded image → NDVI estimate → updates zone μ and σ, re-runs the engine.

Grounded Gemini copilot
Persona- and language-aware answers, temperature 0.2, guardrails, offline fallback.

Persona views, voice & sandbox
Executive audit export, agronomist prior/posterior, farmer canal-gate schedule; Web Speech voice; budget 1,800–5,000 kL and K = 1–3 sliders.

Process flow diagram



Re-run loop: changing the budget or sensor quota in the sandbox, uploading a new canopy image, switching basin, or pressing "Run Decision Engine" re-executes steps 3–7 for all four policies. A full four-policy run took ≈0.6 ms on average in our benchmark, so the dashboard updates instantly.

Wireframes/Mock diagrams of the proposed solution

Watershed Director
Executive macro view

- KPI scorecard (loss, Jain, compliance, VOI)
- Allocation vs. quota bar
- Governance invariants
- Export audit log (JSON)

Field Agronomist
Telemetry & sensors

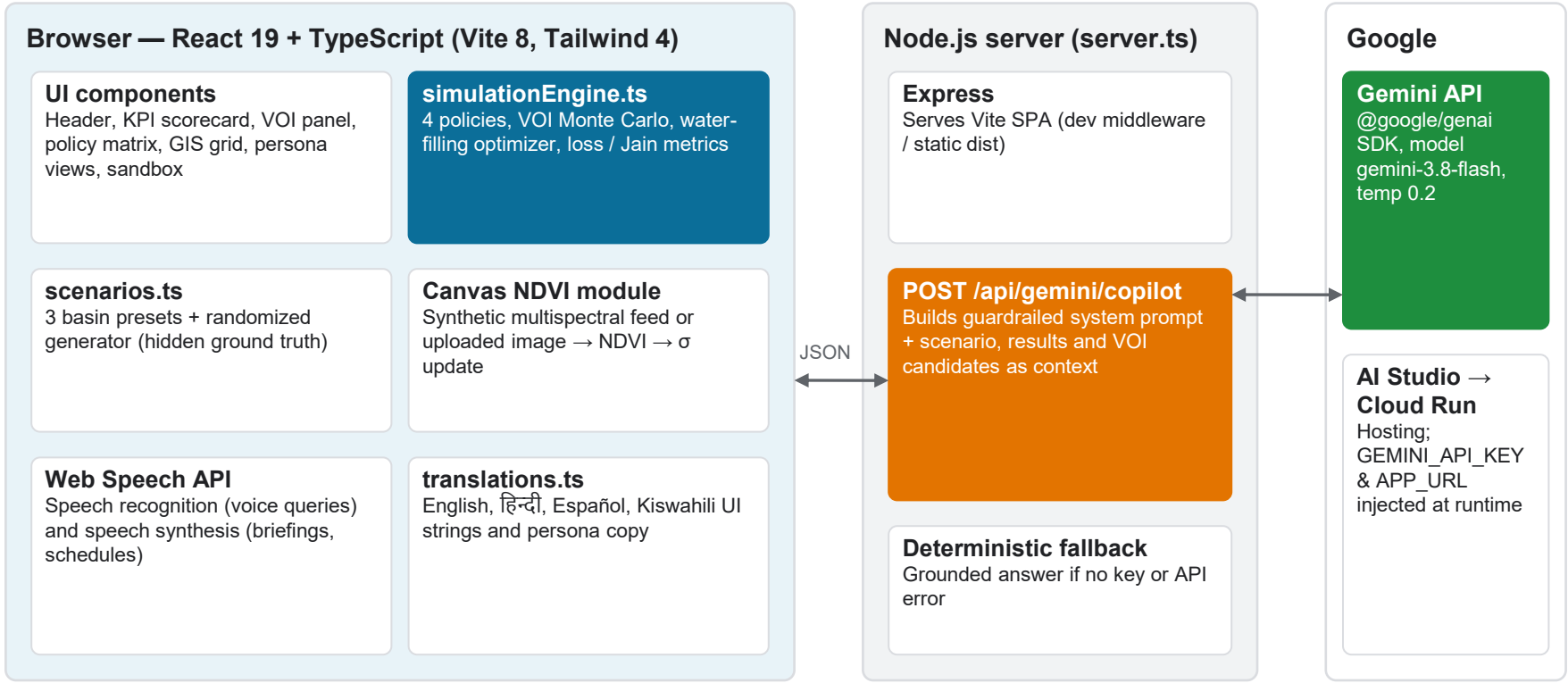
- Probed zone: prior vs. posterior
- Gaussian density chart (95% CI)
- Battery & canopy NDVI
- Drone NDVI scanner

Farmer Co-op Lead
Actionable schedules

- Per-zone delivery volume (kL)
- Canal-gate day & time window
- Optimal / conservation mode
- Play voice schedule

Shared across all three: header with basin selector, persona switch, language (EN/HI/ES/SW), voice briefing; four-policy matrix; GIS basin grid; copilot drawer.

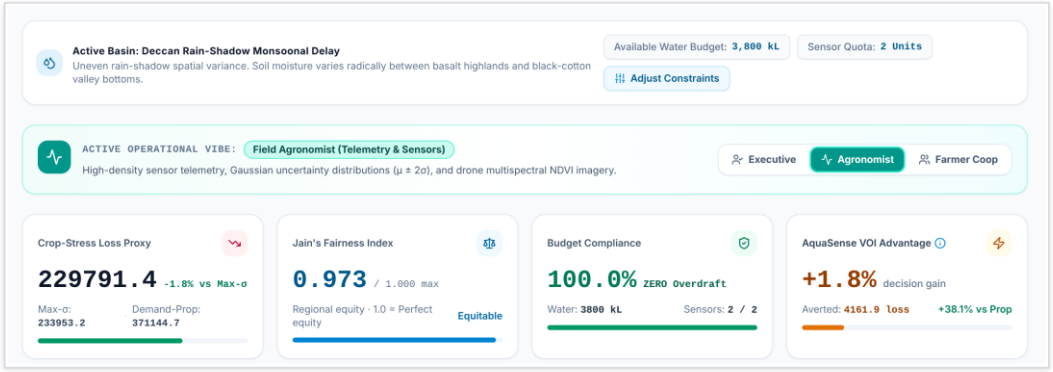
Architecture diagram of the proposed solution



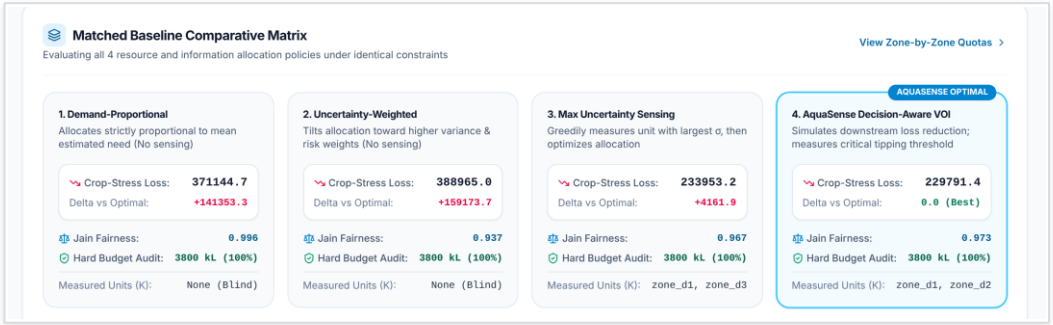
Technologies to be used in the solution

Google AI	Gemini API through the @google/genai SDK (gemini-3.8-flash, temperature 0.2) for the grounded copilot; app built and hosted with Google AI Studio on Cloud Run
Frontend	React 19, TypeScript, Vite 8, Tailwind CSS 4, Motion (animation), lucide-react (icons), HTML Canvas (NDVI imagery & charts as SVG)
Backend	Node.js, Express 4, tsx runtime, dotenv; single /api/gemini/copilot endpoint with deterministic fallback
Decision science	Bayesian-style belief update, Monte-Carlo Value of Information (Box–Muller normal sampling), convex water-filling allocation (bisection on Lagrange multiplier), Jain's fairness index
Voice & language	Web Speech API (SpeechRecognition + SpeechSynthesis) in en-US, hi-IN, es-ES, sw-KE; full UI translations
Tooling	GitHub, TypeScript type-checking (tsc --noEmit), Vite production build

Snapshots of the prototype — dashboard & policy matrix



KPI scorecard (Deccan scenario, Agronomist view)

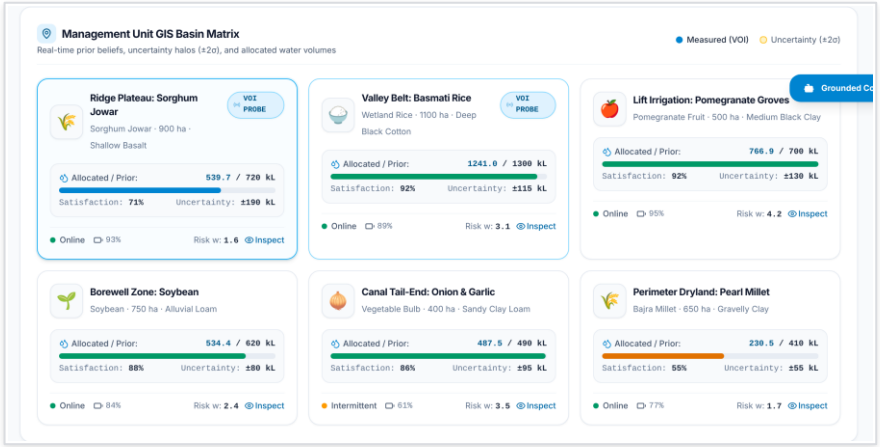


Four policies on identical constraints; AquaSense VOI lowest loss in this run

What this run shows

- Water budget 3,800 kL, K = 2 sensors
- VOI measured Ridge Plateau + Valley Belt; max-σ heuristic measured Ridge Plateau + Pomegranate
- Loss proxy 229,791 vs. 233,953 (−1.8%) vs. 371,145 demand-proportional
- Jain index 0.973; 100% budget compliance

Snapshots of the prototype — basin grid & drone NDVI scanner



GIS basin grid: allocation vs. prior, ±σ, telemetry

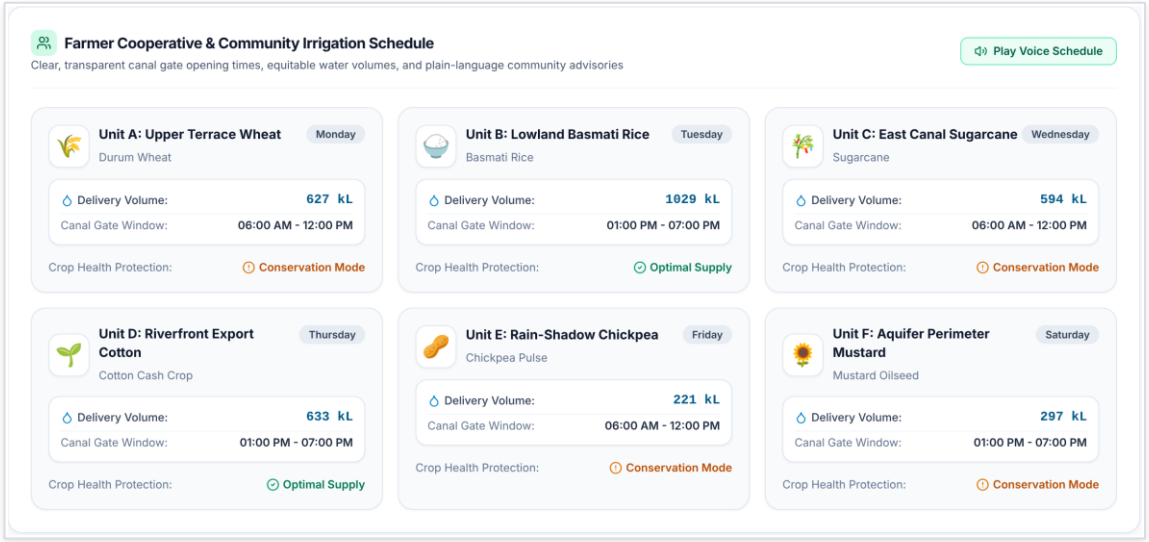


NDVI canopy scanner feeding zone uncertainty

Each zone card shows the allocated volume against its prior need, satisfaction %, uncertainty band, telemetry and battery. Zones picked by the VOI engine carry a "VOI PROBE" badge.

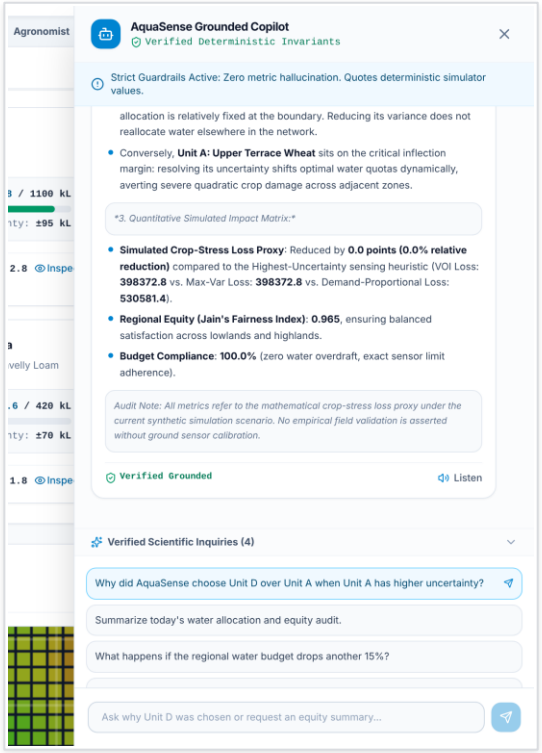
The scanner renders a synthetic multispectral feed (healthy / moderate / severe) or estimates NDVI from an uploaded image, then pushes the result into the selected zone's μ and σ and re-runs all four policies.

Snapshots of the prototype — farmer schedule & grounded copilot



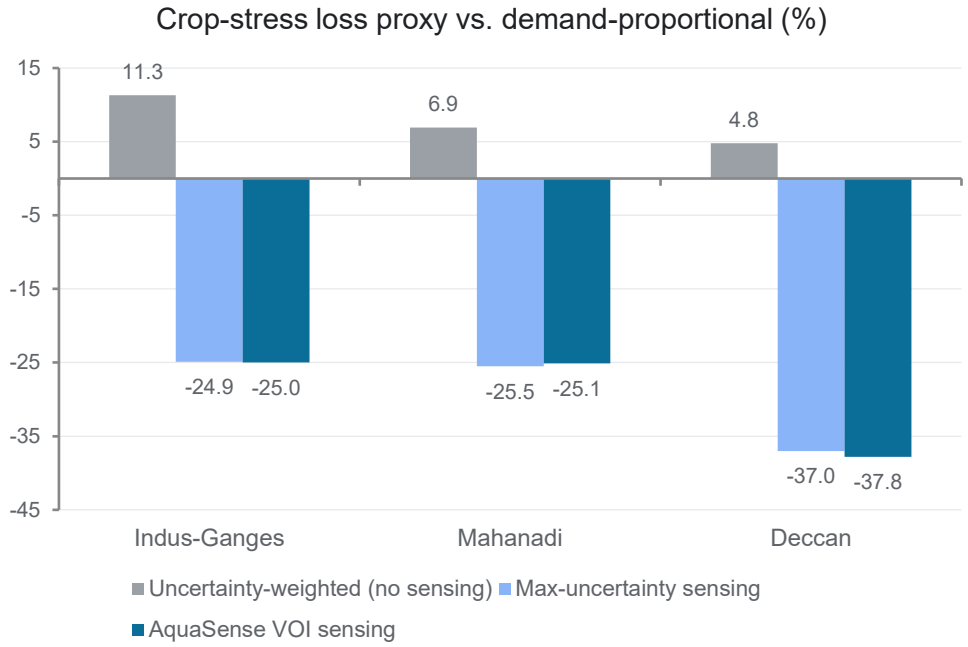
Farmer Co-op view: delivery volume, canal-gate window and mode per zone

Copilot replies always label results as a simulated crop-stress loss proxy under a synthetic scenario and quote only engine output; the "Verified Grounded" tag shows the answer source.



Copilot answering from simulator numbers

Prototype Performance report/Benchmarking



Lower is better. Mean of 1,000 runs per scenario; loss = Σ w·max(0, true need − allocation)².

100% budget compliance in all 16,000 policy evaluations (0 kL overdraft)

25–38% lower loss proxy from sensing + optimizer vs. demand-proportional

≈0.6 ms per full four-policy simulation (Node, average of 4,000 runs)

VOI vs. max-uncertainty, head to head
Deccan: VOI better in 76% of runs (median −1.8% loss).
Indus-Ganges: same zone picked in 94% of runs (tie).
Mahanadi: VOI 0.5% worse on average. Demand-proportional keeps the highest Jain index (≈0.995).

Additional Details/Future Development

System-wide VOI scoring

Score each candidate by total network loss across all zones (today only the measured zone's own loss is scored, so picks often coincide with the max-σ heuristic); more Monte-Carlo samples and a seeded RNG for reproducible picks.

Real data in the loop

Ingest soil-moisture IoT telemetry, Sentinel-2 imagery and weather forecasts; send drone/satellite images to Gemini's multimodal models instead of the current RGB green-ratio NDVI proxy.

Enforce equity in the optimizer

Turn the stated 45% survival-irrigation floor for food-staple zones into a hard optimizer constraint and expose the equity–loss trade-off as a slider.

Field calibration & pilots

Calibrate crop risk weights and water-need priors with a water-user association; validate against measured yield before claiming real-world savings.

Season-long scheduling

Multi-period allocation with canal hydraulics and reservoir inflow forecasts; SMS/IVR delivery of gate schedules in local languages.

AquaSense Project Links

GitHub Public Repository
github.com/kartik-batta/aquasense

Presentation (PDF)
github.com/kartik-batta/aquasense/blob/main/docs/media/aquasense-presentation.pdf

Demo Video Link (3 Minutes)
github.com/kartik-batta/aquasense/blob/main/docs/media/aquasense-demo.mp4

Final Product Link
aquasense.ai.studio

Google Cloud H2S

AI Builder Cup

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Thank you

